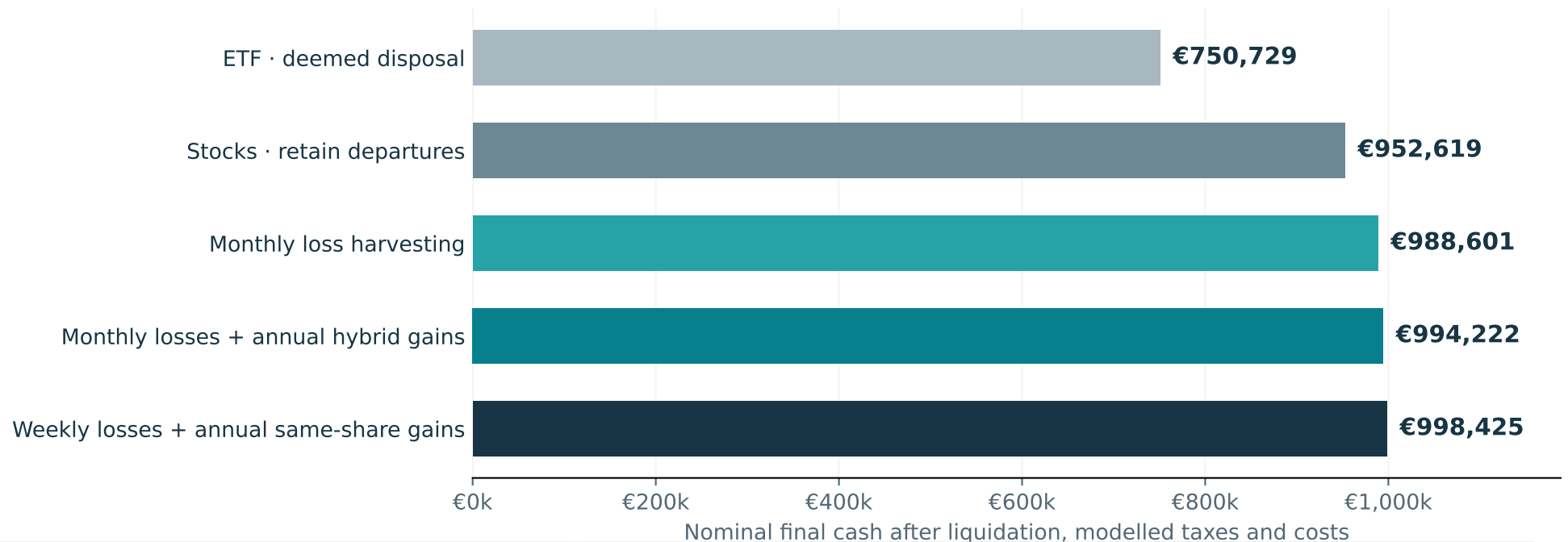


Nasdaq-100, after Irish tax

Starts at €1,000/month · Irish CPI each September · €210,936 contributed



Scenario	Total modelled tax	Trading costs	Wealth vs stock baseline
ETF	€330,840	€0	€-201,890
Stock baseline	€375,605	€2,357	€+0
Monthly losses	€395,436	€3,091	€+35,982
Monthly + hybrid gains	€388,330	€3,552	€+41,603
Weekly + same-share gains	€391,460	€3,226	€+45,806

Highest selected result: Weekly losses + annual same-share gains (€998,425). Every strategy uses the same CPI-indexed contributions. Results are nominal, after final liquidation. The stock baseline retains departures without discretionary harvesting; differences include changed investments as well as tax.

Measure each change against its own control

Effects measured against different controls should not be added together

Feature / existing rules	After-tax wealth change	CGT reduction	Extra nominal exemption relief	Extra trading costs
Departure management No voluntary harvesting	€+27,380	€-13,392	€+163	€+220
Loss harvesting Monthly; annual gains off	€+8,602	€-6,985	€-1,778	€+514
Annual gain harvesting Monthly losses; departed first	€+5,621	€+6,429	€+6,366	€+461
Loss harvesting Weekly; annual gains off	€+9,107	€-7,249	€-1,778	€+401
Annual gain harvesting Weekly losses; restore shares	€+9,319	€+4,352	€+5,948	€+248
Annual exemption Weekly combined strategy	€+13,243	€+3,111	€+6,368	€+224
Annual exemption Monthly hybrid strategy	€-700	€+10,134	€+6,787	€+386

Wealth change compares complete replays. CGT reduction is the difference in modelled CGT paid; changed investments can also change taxable returns. Nominal exemption relief is 33% of the difference in exempt gains on the two ledgers. Neither number is a separate bonus to add to final wealth.

Exemption sensitivities switch the allowance off/on; gain-harvesting comparisons change trading rules. In the weekly case, the allowance changes full-strategy wealth by €13,243, but nominal exemption relief is €6,368. Tax capacity changes later holdings and cash decisions: the wealth change is not an allowance tax saving. Four interaction-control pairs remain in `marginal_effects.csv`.

Interaction between loss and gain harvesting: Weekly losses / annual gains: €+5,841; Monthly losses / annual hybrid gains: €+3,583. These measure how the loss-harvesting effect changes when gain harvesting is enabled. Standalone effects cannot simply be combined; differences along a consistent sequence of controls do sum correctly.

The rules, and the relief actually used

All three selected stock strategies review departed holdings quarterly

Selected strategy	Loss and gain review schedule
Monthly loss harvesting	Monthly losses; no annual gain review.
Monthly + hybrid gains	Monthly losses; December gains sell departed holdings first, then restore current shares.
Weekly + same-share gains	Friday loss reviews; December gains sell and repurchase eligible current shares.

Strategy	Exempt gains before final year	Earlier years using full allowance	Nominal relief including final year	Net losses from TLH disposals
Monthly losses	€5	0	€421	€28,356
Monthly + hybrid gains	€19,296	14	€6,787	€34,712
Weekly + same-share gains	€18,028	14	€6,368	€31,471

Loss review: sell an eligible whole holding when its euro loss after fees reaches both 5% and €25. Require no same-class purchase in the preceding 29 days and block repurchase for the next 29 days. Reinvest into eligible current underweights; every strategy shares the CPI-indexed contribution schedule.

Departure review: sell confirmed former constituents only when existing realised losses, carried losses and the remaining exemption cover the gain without increasing CGT at that review. A profitable departure can therefore remain in the portfolio.

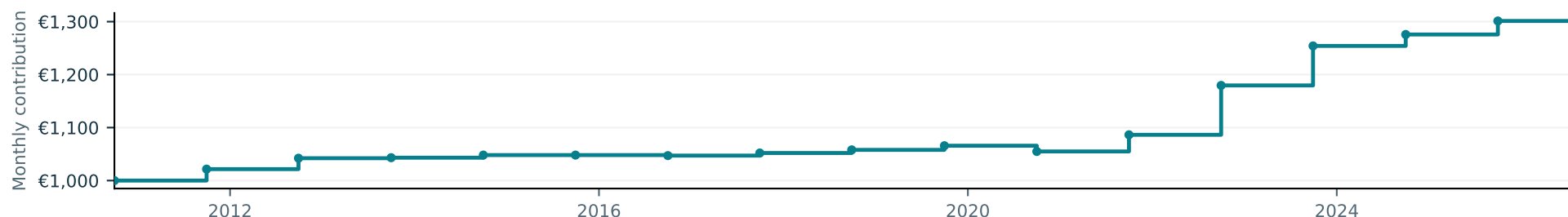
Annual gain review: partial FIFO sales use losses before the €1,270 allowance. Same-share mode ranks by gain per euro sold; hybrid ranks departures then current overweights. Immediate replacement requires non-losing selected lots, current membership and the purchase restriction; both legs pay costs. Hybrid departure sales fund current underweights.

Harvested losses are not permanent relief equal to loss × 33%: replacement shares can have lower cost basis and a larger later gain. This study touches 17 tax years, including partial 2010 and final-liquidation 2026. All figures are hypothetical; no unrelated gains consume the exemption.

Contributions, evidence and practical limits

Annual Irish CPI adjustments use information available at the time; all reported wealth is nominal

Monthly amount: €1,000.00 initially → €1,301.28 finally | 15 anniversary reviews | 192 contributions



CASHFLOWS • Each September: $\text{€1,000.00} \times \text{August CPI} / \text{August 2010 CPI}$, rounded to cents and fixed for 12 payments. Both inflation and deflation apply. CSO all-items CPI uses December 2006=100 and releases predating each review; a current snapshot relies on CSO's non-revision policy. Identical scheduled amounts fund both cases; three stock payments post next trading session. There is €18,936 more contributed than keeping the initial monthly amount fixed; wealth remains nominal.

PERIOD AND INPUTS • 30 September 2010 to 30 September 2026. Stock prices extend through the final session; ETF/FX observations through 2 October. Last usable weights: 31 August, available 7 September. September weights were unavailable until 7 October. Membership, weights, dividends and corporate actions remain provisional; some unquoted holdings use nontradable valuation marks.

SELECTION • Strategies were selected retrospectively from the earlier fixed-contribution exploration. Causal trade rules do not make that selection an out-of-sample test. CPI-linked cashflows can change holdings, tax capacity, triggers and rankings. Matched controls describe this replay, not a universally best strategy.

FROZEN TAX AND COSTS • 33% share CGT; €1,270 annual exemption; 52.35% dividend tax; 38% ETF fund tax with eight-year deemed disposal. Losses precede the exemption; withholding credits are included. Non-euro share trades incur 0.15% FX per side; no extra spread or slippage is modelled. ETF NAV embeds fund costs. Assumed rates are not each year's law or anyone's actual tax status.

INTERPRETATION • Share matching, foreign reorganisations and compulsory receipts retain documented assumptions. CGT reserves and annual settlements approximate payment timing. Audits check arithmetic and trade eligibility, not tax rulings, historical broker availability or executable fills.

REPRODUCE LOCALLY • Run `python historical/run_tax_optimization_study.py --workers 3` with the required inputs, then `python historical/run_tax_optimization_report.py`. Current tables, `contribution_schedule.csv`, ledgers and input/code hashes are in `historical/results/latest/`. The schedule records CPI evidence and availability; the manifest documents its method.

Read the assumptions before interpreting small differences as an investable advantage.

Research only, not tax, legal or investment advice. See the README legal notice.

[Revenue](#) · [CGT calculation and exemption](#)

[Revenue](#) · [Share matching restrictions](#)

Previously weak rules, tested again

Selected from the earlier fixed-contribution study; this replay uses matched CPI-indexed cashflows

Control A (€995,121): monthly loss harvesting at 5% and €25, quarterly tax-budget departure reviews and annual same-share gain harvesting. Control B (€983,476): the same departure and gain rules, without discretionary loss harvesting. Results include final liquidation, taxes and costs.

Change / matched comparator	Final cash	Wealth change vs control	Extra trading costs	CGT reduction vs control
Harvest any monthly loss $\geq$ €1 Control A	€972,752	€-22,369	€+741	€+10,235
Retain departures; gains only Control B	€956,852	€-26,624	€-236	€+13,261
Review gains every month Control A	€957,621	€-37,500	€+133	€+18,351

HARVEST ANY MONTHLY LOSS $\geq$ €1 • Relaxing the loss threshold changes loss-sale eligibility, replacement holdings and purchase restrictions. The matched comparison measures the resulting wealth, tax and cost effects; a larger harvested loss is not itself a cash benefit.

RETAIN DEPARTURES; GAINS ONLY • Retaining departed stocks prevents tax-budget exits and reinvestment of those proceeds into current constituents. Differences in wealth reflect the retained exposure as well as taxes and trading costs. Versus the plain stock baseline: €+4,234.

REVIEW GAINS EVERY MONTH • Monthly gain reviews change disposal timing, cost-basis resets, later loss eligibility and holdings. An annual allowance is not multiplied by monthly reviews; the matched replay measures the combined wealth effect.

Ledger evidence	Loss sales variant / control	Gain sales variant / control	Net TLH losses realised	Extra nominal exemption relief
Harvest any monthly loss $\geq$ €1	1,403 / 424	268 / 198	€37,082	€-120
Retain departures; gains only	0 / 0	143 / 139	€0	€-2
Review gains every month	453 / 424	720 / 198	€37,506	€-80

Selection from the earlier study does not predetermine this replay's ranking. Fees, holdings, tax basis and later decisions all change; the comparisons do not separately attribute each mechanism. Lower tax can also reflect lower investment gains.